

Architecture Selection & Technical Justification

System: Second-Hand Product Escrow Marketplace (Problem Statement #39) | Student: Atishay Jain (PES1UG24CS097)

Architecture Selection: “We chose Layered Architecture with Domain Service Segregation for the Second-Hand Product Escrow Marketplace System.”

1. Architectural Choice

We selected a three-tier **Layered Architecture** structured into horizontal tiers: **Presentation Layer** (responsive web and mobile client applications for buyers, sellers, inspectors, and dispute officers), **Business Layer** (orchestrating decoupled domain components: Product Catalog & Listing, Order & Escrow Manager, Physical Inspection Service, Payment Gateway Adapter, and Dispute Arbitration), and **Data & Compliance Layer** (transactional relational persistence alongside an immutable, append-only cryptographic audit store). Unidirectional downward dependencies guarantee that external requests cannot bypass domain policies.

2. Two Scenario-Related Reasons for Selection

- **Financial Invariant Enforcement & Escrow Integrity (FR-001, FR-004):** In an escrow marketplace, transactions must satisfy strict preconditions: buyer funds remain locked under state “Under Inspection” and cannot release to the seller until both an authorized inspection sign-off (FR-001) and a buyer delivery confirmation (FR-004) are registered. Layered encapsulation strictly forbids presentation clients from directly triggering payout transfers or updating ledger balances; all state transitions must be authorized and validated sequentially by the central Order & Escrow Manager Component, eliminating premature disbursements and fraud vectors.
- **Domain Decoupling & Independent Operational Lifecycles (FR-002, FR-003, FR-005):** The platform unites three radically distinct user journeys: public catalog exploration (high-volume read browsing), on-site physical inspection (photo uploads and condition certifications by independent agents), and formal legal arbitration (dispute case routing). Decoupling these capabilities into dedicated components behind clean UML interfaces (IProductCatalog, IInspectionReport, IDisputeArbitration) allows each service to scale, deploy, and undergo isolated code maintenance without risking regression in the mission-critical escrow core.

3. Security Advantage: Defense-in-Depth & Tamper-Evident Audit Logging (NFR-002)

The layered model establishes an impenetrable **defense-in-depth perimeter**. The presentation tier acts as a hardened API gateway executing input sanitization, token validation, and rate-limiting, preventing untrusted client payloads from reaching core services. Payment gateway credentials and financial authorization routines are isolated exclusively within the Payment Gateway Adapter Component in the private subnet. Furthermore, this architecture natively supports an **Immutable Audit Ledger Component** (NFR-002): every order status transition, inspector sign-off, delivery confirmation, and dispute freeze is intercepted and written to a SHA-256 hash-chained, append-only journal with 7-year retention and TLS 1.3/AES-256 encryption, ensuring legally admissible, tamper-evident non-repudiation during arbitration.

4. Performance Benefit: Sub-Second Dispute Freezing & Asynchronous Scaling (NFR-001)

Per NFR-001, filing a dispute must freeze escrow funds immediately to prevent capital flight and fraudulent account withdrawals. Our architecture isolates high-priority transactional commands from read-heavy catalog traffic. When a buyer submits a dispute, the Order Manager and Dispute Arbitration Component execute a dedicated, lightweight **row-level write lock** on the escrow account in under **200 ms**, easily surpassing the strict 1-second benchmark. Simultaneously, browse/search operations (UC-02) are offloaded to read replicas and in-memory caches, guaranteeing snappy UI responsiveness under peak marketplace concurrency.